

RetinAnalysis — New Machine Setup Guide

Everything to download and install to run this repo on Windows or Mac

August 2026

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Before You Start

This repo (`retinanalysis`) is a heavily customized fork — the analysis code, demo notebooks, and several utility files have significant lab-specific changes beyond the original upstream project. None of that requires any extra setup step beyond what's below: everything custom lives inside this one repository (plus one required submodule), so a normal `git clone` of the up-to-date repo already

includes all of it. Push your current work to GitHub first (`git add`, `git commit`, `git push`) so a fresh clone actually picks up everything described here.

Everything you'll install, at a glance:

Tool	Why you need it	Windows	Mac
Git	Clone the repo (and its submodule)	Git for Windows	Xcode Command Line Tools or brew install git
Docker Desktop	Runs the local MySQL/DataJoint database	Requires WSL2	Get the right build for your chip
A C/C++ compiler	Needed to build the vision-utils submodule (Cython + pybind11 extensions)	Visual Studio Build Tools	Xcode Command Line Tools
Python 3.10–3.12	Runs retinanalysis itself	via uv or Miniconda	via uv or Miniconda
uv or conda	Creates the Python environment and installs packages	pick one	pick one

Everything below assumes a fresh machine with none of this installed yet — skip anything you already have.

Step 1 — Install Git

Mac: Git ships with the Xcode Command Line Tools. Open Terminal and run:

```
xcode-select --install
```

This also installs **clang**, the C/C++ compiler needed in Step 3, so it does double duty — install it now.

Windows: Download and run the installer from <https://git-scm.com/download/win>. Accept the defaults. This also installs **Git Bash**, a bash-compatible terminal — you'll want it in Step 5, since the repo's installer script (`install.sh`) is a bash script and won't run in plain PowerShell or Command Prompt.

Step 2 — Install Docker Desktop

The local database (MySQL, via DataJoint) runs in a Docker container, so Docker Desktop is required for any database-backed workflow (queries, `ra.populate_database()`, etc.). `import retinanalysis` itself does **not** need Docker running.

Download from <https://docs.docker.com/desktop/>.

Windows:

- Requires Windows 10 64-bit (version 21H2+) or Windows 11, with hardware virtualization enabled in the BIOS/UEFI, and an admin account.
- Docker Desktop uses the WSL2 backend. The installer will offer to enable and install WSL2 for you if it isn't already set up — accept that, then restart when prompted.

Mac:

- Download the build that matches your chip — Apple Silicon (M1/M2/M3/M4) or Intel. The wrong build will refuse to launch.
- Supported on macOS Ventura (13) through the current release.

Both platforms: plan for roughly 6 GB of free disk space (Docker Desktop itself plus the Linux VM/image it manages), 4 GB RAM minimum (8 GB+ recommended).

After installing, **launch Docker Desktop at least once** and leave it running whenever you need database access — you'll come back to it in Step 7.

Step 3 — Install a C/C++ Compiler

The `vision-utils` package (a required submodule, used to read Vision's `.ei/.bin` recording files) compiles Cython and pybind11 C++ extensions at install time. This needs an actual C/C++ compiler present on the machine — pure `pip install` will fail without one.

Mac: Already covered in Step 1 (`xcode-select --install` installs `clang`). Nothing further needed.

Windows: Install **Microsoft C++ Build Tools**:

1. Download the Visual Studio Build Tools installer from <https://visualstudio.microsoft.com/visual-cpp-build-tools/>.
2. In the installer, select the “**Desktop development with C++**” workload.
3. Install (this is a few GB) and restart if prompted.

You do not need full Visual Studio or an IDE — the Build Tools package alone is enough.

Step 4 — Install a Python Environment Manager

`retinanalysis` requires Python 3.10–3.12 (the repo's installer defaults to 3.11.13). Pick **either** `uv` or `conda` — you don't need both. `uv` is faster and is what the repo's own README leads with; `conda` is a fine, equally supported alternative if your lab already standardizes on it.

Option A — `uv` (recommended)

Mac/Linux, in Terminal:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Windows, in PowerShell:

```
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

Open a **new** terminal window afterward so it picks up the updated PATH, then confirm with `uv --version`.

Option B — conda (Miniconda)

Download the installer for your OS/chip from <https://docs.conda.io/en/latest/miniconda.html> and run it, accepting the defaults. Confirm with `conda --version` in a new terminal.

Step 5 — Clone and Install RetinAnalysis

Use a terminal that has `git` on its PATH — on Windows, use **Git Bash** (installed in Step 1) so the bash installer script works.

```
git clone https://github.com/DRezeanu/retinanalysis.git --recursive
cd retinanalysis
```

`--recursive` is required — it pulls in the `artificial-retina-software-pipeline` submodule (`lib/`) that `vision-utils` lives in. If you forget it, fix it with:

```
git submodule update --init --recursive
```

Then run the installer with whichever tool you picked in Step 4:

```
./install.sh uv
# or:
./install.sh conda
```

This creates the Python environment, installs `retinanalysis` itself (editable), builds and installs the `vision-utils` submodule package, and creates a placeholder `src/retinanalysis/config/config.ini` if one doesn't already exist. Useful variants:

```
./install.sh uv --dev           # also installs pytest
./install.sh conda --env my_env  # custom conda environment name
./install.sh uv --python 3.11.13 # pin a specific Python version
```

If **Git Bash** / a bash shell genuinely isn't available on Windows, you can run the same steps `install.sh` performs by hand, from PowerShell, inside your activated environment:

```
pip install -e .
pip install .\lib\artificial-retina-software-pipeline\utilities\
```

(swap `pip` for `uv pip` if using a `uv` venv). You'll then need to create `src\retinanalysis\config\config.ini` yourself — see Step 6.

Step 6 — Fix a Known Windows Matplotlib Bug

On Windows, the first `import matplotlib` (which happens indirectly via `import retinanalysis`) can raise a DLL load error. If you hit this, run:

```
pip uninstall Pillow
pip install -U Pillow
```

(`uv pip uninstall Pillow` / `uv pip install -U Pillow` if using a `uv` environment.)

Step 7 — Configure config.ini

`install.sh` creates a placeholder at `src/retinanalysis/config/config.ini` (or create it yourself per Step 5’s manual path). Edit it with real data paths for **this** machine — it’s gitignored, so every machine keeps its own copy and it’s never overwritten by `git pull`.

```
[DEFAULT]
analysis = /path/to/analysis
data = /path/to/sorted
raw = /path/to/raw
h5 = /path/to/h5
meta = /path/to/meta
tags = /path/to/tags
query = /path/to/query/analysis
user = your_username

[SECONDARY]
analysis = /path/to/secondary/analysis
data = /path/to/secondary/sorted
raw = /path/to/secondary/raw
h5 = /path/to/secondary/h5
meta = /path/to/secondary/meta
tags = /path/to/secondary/tags
query = /path/to/secondary/query/analysis
user = your_username

[LINUX_DEFAULT]
...
[LINUX_SECONDARY]
...
[WINDOWS_DEFAULT]
...
[WINDOWS_SECONDARY]
...
```

The package auto-selects the `WINDOWS_*` / `LINUX_*` / plain (macOS) sections based on the OS it’s running on, so fill in whichever section(s) apply to this machine — you don’t need every section populated, just the ones this machine will actually use. The `query` path is used for plotting mosaics/cell typing across the whole dataset archive (e.g. a NAS mount) even when your other paths point at a local SSD.

Step 8 — Start the Local Database

Retinanalysis stores experiment metadata in a local DataJoint/MySQL database, run via Docker.

```
mkdir -p ../retinanalysis-database
cp docker-compose.yaml ../retinanalysis-database/
cd ../retinanalysis-database
docker compose up -d
```

(Older Docker installs may need the hyphenated `docker-compose up -d` instead.)

Make sure **Docker Desktop is open and the container is running** before any database-backed call — in Docker Desktop’s UI you’ll see a stop icon next to the running container; click the play button if it isn’t running.

Step 9 — Populate the Database

From your installed Python environment:

```
import retinanalysis as ra
ra.populate_database()
```

If `config.ini` is set up correctly, this needs no arguments. `import retinanalysis` itself works without the database running — only queries and `populate_database()` actually need the container up.

Optional — `datajoint.json`

DataJoint 2 will print a harmless warning if it can’t find a `datajoint.json` in your project root. Retinanalysis has working fallback defaults built in, so this is optional. To silence the warning explicitly, create `datajoint.json` in the repo root:

```
{
  "database.host": "127.0.0.1",
  "database.port": 3306,
  "database.user": "root",
  "database.password": "simple"
}
```

Quick Reference — Full Command Sequence

Once every prerequisite above is installed, a fresh setup is just:

```
git clone https://github.com/DRezeanu/retinanalysis.git --recursive
cd retinanalysis
./install.sh uv                # or: ./install.sh conda
# edit src/retinanalysis/config/config.ini with real paths
mkdir -p ../retinanalysis-database
cp docker-compose.yaml ../retinanalysis-database/
cd ../retinanalysis-database
docker compose up -d
```

```
import retinanalysis as ra
ra.populate_database()
```

Troubleshooting

“DLL load failed” on `import matplotlib` (Windows) — see Step 6 (reinstall Pillow).

`install.sh` fails with “required submodule package is missing” — you cloned without `--recursive`. Run `git submodule update --init --recursive` from the repo root, then rerun `install.sh`.

Docker Desktop won’t start / WSL2 errors (Windows) — confirm hardware virtualization is enabled in BIOS/UEFI, and that WSL2 itself is installed (`wsl --status` in PowerShell). Docker Desktop’s installer can install WSL2 for you if it’s missing.

Database calls fail / connection refused — Docker Desktop isn’t running, or the `retinanalysis-database` container isn’t started. Open Docker Desktop and confirm the container shows as running (stop icon, not play icon), or run `docker compose up -d` again from the `retinanalysis-database` directory.

Using an existing pre-DataJoint-2 database — don’t try to reuse it. Create a fresh container per Step 8 and repopulate per Step 9 after updating `retinanalysis` — mixing an old DataJoint 0.14 database with the current `datajoint==2.2.2` dependency isn’t supported. Keep the old environment/database around until the new one is confirmed working.

pip/uv install of vision-utils fails with a compiler error — the C/C++ compiler from Step 3 either isn’t installed or isn’t on PATH. On Windows, make sure you installed the “Desktop development with C++” workload, not just the base Build Tools installer, and try again from a fresh terminal.

Sources

- Docker Desktop system requirements: <https://docs.docker.com/desktop/>
- uv installation: <https://docs.astral.sh/uv/getting-started/installation/>
- Miniconda: <https://docs.conda.io/en/latest/miniconda.html>
- Git for Windows: <https://git-scm.com/download/win>
- Microsoft C++ Build Tools: <https://visualstudio.microsoft.com/visual-cpp-build-tools/>